**Problem 10.33**

Which of the following are true?

$$\text{(I)} \quad \frac{d}{dd}(i) = v^{-2},$$

$$\text{(II)} \quad \frac{d}{di}\left(i^{(m)}\right) = v^{-\left(\frac{m-1}{m}\right)},$$

$$\text{(III)} \quad \frac{d}{d\delta}(i) = 1+i.$$

**Solution**

Recall the relationships

$$v = \frac{1}{1+i}, \qquad d = \frac{i}{1+i}, \qquad 1+i = e^{\delta}.$$

**Statement (I).**

Solving the relationship between $d$ and $i$ for $i$ gives

$$i = \frac{d}{1-d}.$$

Therefore,

$$\begin{aligned} \frac{di}{dd} &= \frac{(1-d)+d}{(1-d)^2} \\ &= \frac{1}{(1-d)^2}. \end{aligned}$$

Since $1-d=v$, it follows that

$$\frac{di}{dd} = \frac{1}{v^2} = v^{-2}.$$

Thus, **statement (I) is true**.

**Statement (II).**

The nominal annual interest rate convertible $m$thly satisfies

$$1+i = \left(1+\frac{i^{(m)}}{m}\right)^m.$$

Hence,

$$i^{(m)} = m\left[(1+i)^{1/m} - 1\right].$$

Differentiating with respect to $i$,

$$\begin{aligned} \frac{d}{di}\left(i^{(m)}\right) &= m\left[\frac{1}{m}(1+i)^{1/m-1}\right] \\ &= (1+i)^{-\left(\frac{m-1}{m}\right)} \\ &= v^{\frac{m-1}{m}}. \end{aligned}$$

This is not $v^{-(m-1)/m}$. Thus, **statement (II) is false**.

**Statement (III).**

From $1+i=e^{\delta}$, we have

$$i = e^{\delta} - 1.$$

Consequently,

$$\frac{di}{d\delta} = e^{\delta} = 1 + i.$$

Thus, **statement (III) is true**.

Therefore, the correct answer is

$$\boxed{\text{Statements (I) and (III) are true.}}$$